

Moments

In [mathematics](#), the **moments** of a [function](#) are quantitative measures related to the shape of the function's [graph](#). The concept is used in both [mechanics](#) and [statistics](#). If the function represents mass, then the zeroth moment is the total [mass](#), the first moment divided by the total mass is the [center of mass](#), and the second moment is the [rotational inertia](#). If the function is a [probability distribution](#), then the zeroth moment is the total probability (i.e. [one](#)), the first moment is the [expected value](#), the second [central moment](#) is the [variance](#), the third [standardized moment](#) is the [skewness](#), and the fourth standardized moment is the [kurtosis](#). The mathematical concept is closely related to the concept of [moment](#) in physics.

Moments can be defined as the arithmetic mean of various powers of deviations taken from the mean of a distribution. These moments are known as central moments. The first four moments about arithmetic mean or central moments are defined below.

	Ungrouped data	Grouped data
First moments about the mean; ₁	$\frac{\sum (X - \bar{X})}{n} = 0$	$\frac{\sum f(X - \bar{X})}{N} = 0$
Second moments about the mean; ₂	$\frac{\sum (X - \bar{X})^2}{n} = \sigma^2$ (variance)	$\frac{\sum f(X - \bar{X})^2}{N} = \sigma^2$
Third moments about the mean ; ₃	$\frac{\sum (X - \bar{X})^3}{n}$	$\frac{\sum f(X - \bar{X})^3}{N}$
Fourth moment about the Mean ; ₄	$\frac{\sum (X - \bar{X})^4}{n}$	$\frac{\sum f(X - \bar{X})^4}{N}$

is a Greek letter, pronounced as 'mu'].

Moments about a constant value are (for ungrouped data)

$$\mu_1' = \frac{\sum (X - A)}{n}, \mu_2' = \frac{\sum (X - A)^2}{n}, \mu_3' = \frac{\sum (X - A)^3}{n}, \mu_4' = \frac{\sum (X - A)^4}{n}$$

For grouped data:

$$\mu_1' = \frac{\sum f(X-A)}{N}, \mu_2' = \frac{\sum f(X-A)^2}{N}, \mu_3' = \frac{\sum f(X-A)^3}{N}, \mu_4' = \frac{\sum f(X-A)^4}{N}$$

In general, given n observations $x_1, x_2, \dots, x_n$ the r^{th} order raw moments ($r=0,1,2,\dots$) are defined as follows:

$$\mu_r' = \frac{1}{N} \sum f(x-A)^r \text{ (about A)}$$

$$\mu_r' = \frac{\sum fx^r}{N} \text{ (about origin)}$$

$$\mu_r = \frac{1}{N} \sum f(x-\bar{x})^r \text{ (about mean)}$$

Relation between raw moments and central moments

$$\mu_1 = 0$$

$$\mu_2 = \mu_2' - (\mu_1')^2$$

$$\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2\mu_1'^3$$

$$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'(\mu_1')^2 - 3(\mu_1')^4$$

Example: The first two moments of the distribution about the value 5 of the variable, are 2 and 20. Find the mean and the variance.

Solution:

$$\mu'_1 = 2, \mu'_2 = 20 \text{ and } A = 5$$

$$\bar{x} = \mu'_1 + A$$

$$\bar{x} = 2 + 5 = 7$$

$$\sigma^2 = \mu'_2 - (\mu'_1)^2$$

$$\sigma^2 = 20 - 2^2 = 16$$

Mean = 7 and Variance = 16

SKWNESS AND KURTOSIS

There are two other comparable characteristics called skewness and kurtosis that help us to understand a distribution.

Skewness

Skewness means ‘**lack of symmetry**’. We study skewness to have an idea about the shape of the curve drawn from the given data. When the data set is not a symmetrical distribution, it is called a skewed distribution and such a distribution could either be positively skewed or negatively skewed.

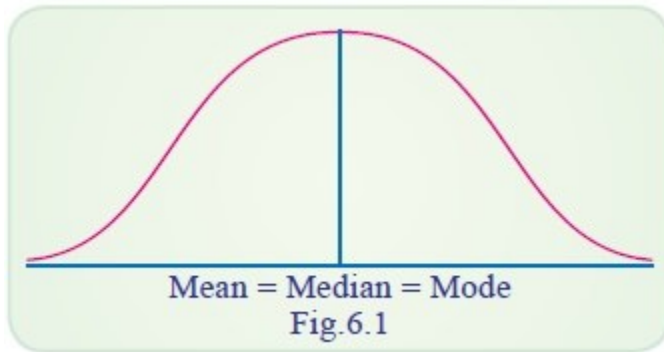


The concept of skewness will be clear from the following three diagrams showing a 1. symmetrical distribution, 2. positively skewed distribution and 3. negatively skewed distribution.

We can see the symmetry from the following diagram.

(a) Symmetrical Distribution

It is clear from the diagram below that in a symmetrical distribution the values of mean, median and mode coincide. The spread of the frequencies is the same on both sides of the centre point of the curve.



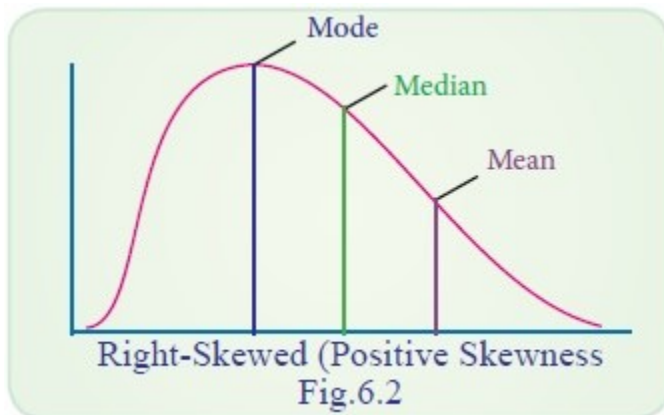
3

7

8

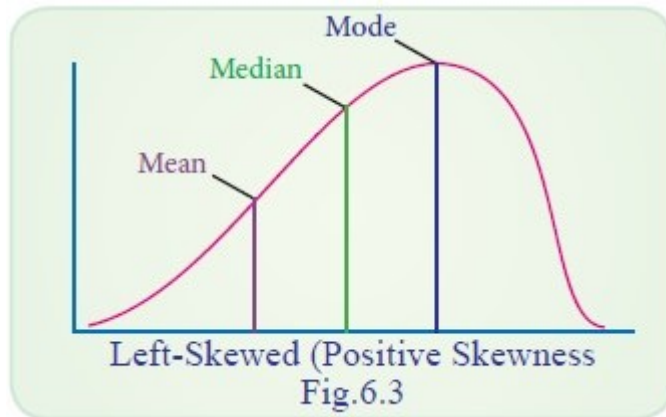
(b) Positively Skewed Distribution

In the positively skewed distribution the value of the mean is maximum and that of mode is least – the median lies in between the two. In the positively skewed distribution the frequencies are spread out over a greater range of values on the high-value end of the curve (the right-hand side) than they are on the low value end. For a positively skewed distribution, $\text{Mean} > \text{Median} > \text{Mode}$



(c) Negatively skewed distribution

In a negatively skewed distribution the value of mode is maximum and that of mean least-the median lies in between the two. In the negatively skewed distribution the position is reversed, i.e., the excess tail is on the left-hand side.



It should be noted that in moderately symmetrical distribution the interval between the mean and the median is approximately one-third of the interval between the mean and the mode. It is this relationship which provides a means of measuring the degree of skewness.

d. Some important Measures of Skewness

(i) Karl-Person coefficient of skewness

(ii) Bowley's coefficient of skewness

(iii) Coefficient of skewness based on moments

(i) Karl-Person coefficient of skewness

$$\text{Karl-Person's Coefficient of Skewness, } SK_p = \frac{\text{Mean} - \text{Mode}}{S.D}$$

If Mean=Mode ,then **Symmetrical Distribution.**

Mean >Mode ,then **Positively Skewed Distribution.**

Mean <Mode ,then **Negatively skewed distribution.**

In case of mode is absent of data

Karl-Pearson Coefficient of skewness , $SK_p = \frac{3(\text{Mean} - \text{Median})}{S.D}$

Mean=Median ,then **Symmetrical Distribution.**

Mean >Median ,then **Positively Skewed Distribution.**

Mean =7.35 median =8

Mean <Median ,then **Negatively skewed distribution.**

Example

From the known data, mean = 7.35, mode=8 and Variance = 1.69 then find the Karl-Pearson coefficient of skewness.

Solution:

$$\text{Karl- Pearson coefficient of skewness} = \frac{\text{Mean} - \text{Mode}}{S.D}$$

$$\text{Variance} = 1.69$$

$$\text{Standard deviation} = \sqrt{1.69} = 1.3$$

$$\begin{aligned}\text{Karl- Pearson coefficient of skewness} &= \frac{7.35 - 8}{1.3} \\ &= \frac{-0.65}{1.3} = -0.5\end{aligned}$$

(ii) Bowley's coefficient of skewness

In Karl Pearson method of measuring skewness the whole of the series is needed. Prof. Bowley has suggested a formula based on position of quartiles. In symmetric distribution quartiles will be equidistance from the median. $Q_2 - Q_1 = Q_3 - Q_2$, but in skewed distributions it may not happen. Hence

$$\text{Bowley's coefficient of skewness (SK)} = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$$

Example

If $Q_1 = 40$, $Q_2 = 50$, $Q_3 = 60$, Bowley's coefficient of skewness

Solution:

Bowley's coefficient of skewness

$$= \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$$

Bowley's coefficient of skewness

$$= \frac{40 + 60 - 2 \times 50}{60 - 40} = \frac{0}{20} = 0$$

$\therefore$ Given distribution is symmetric.

(iii) Measure of skewness based on Moments

The Measure of skewness based on moments is denoted by β_1 and is given by

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}$$

Example

Find β_1 for the following data $\mu_1 = 0$, $\mu_2 = 8.76$, $\mu_3 = -2.91$

Solution:

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}$$

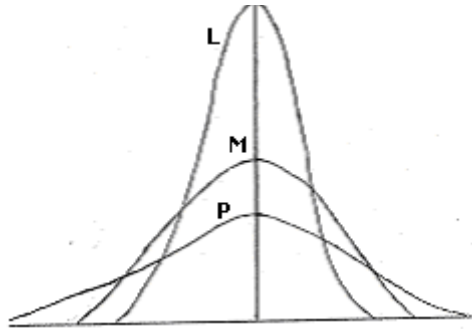
$$\begin{aligned}\beta_1 &= \frac{(-2.91)^2}{(8.76)^3} = \frac{8.47}{672.24} \\ &= 0.0126\end{aligned}$$

Kurtosis

Kurtosis in Greek means ‘bulginess’. In statistics kurtosis refers to the degree of flatness or peakedness in the region about the mode of a frequency curve. The degree of kurtosis of distribution is measured relative to the peakedness of normal curve. In other words, measures of kurtosis tell us the extent of which a distribution is more peaked or flat-topped than the normal curve.

The following diagram illustrates the shape of three different curves mentioned below:

If a curve is more peaked than the normal curve, it is called ‘leptokurtic’. In such a case items are more closely bunched around the mode. On the other hand if a curve is more flat-topped than the normal curve, it is called ‘platykurtic’. The bell shaped normal curve itself is known as ‘mesokurtic’. We can find how much the frequency curve is flatter than the normal curve using measure of kurtosis.



L= Leptokurtic, M=Mesokurtic P= platykurtic

Measures of Kurtosis

The most important measure of kurtosis is the value of the coefficient. It is defined as: coefficient of kurtosis $\beta_2 = \mu_4 / \mu_2^2$

$$\beta_2 = \frac{\mu_4}{\mu_2^2}$$

If $\beta_2 = 3$, the distribution is said to be normal and the curve is mesokurtic.

If $\beta_2 > 3$, the distribution is said to be more peaked and the curve is leptokurtic.

$\beta_2 < 3$, the distribution is said to be flat topped and the curve is platykurtic.

If

Example

Find the value of β_2 for the following data $\mu_1 = 0$, $\mu_2 = 4$, $\mu_3 = 0$, $\mu_4 = 37.6$.

Solution:

$$\beta_2 = \frac{\mu_4}{\mu_2^2}$$

$$\beta_2 = \frac{37.2}{4^2} = \frac{37.6}{16} = 2.35 < 3$$

$\beta_2 < 3$, The curve is platykurtic.

Significance of Skewness, and Kurtosis

Skewness and kurtosis are very important to statistical analysis. Skewness indicate the nature of distributions, i.e., where the mean, median, and mode lie in a numerical information series. Decision madders apply skewness to find their required measures of central value i.e., mean, median or mode. The measure of kurtosis, the fourth device in describing a frequency distribution, can be used to show the degree of concentration, either the values concentrated in the area around the mode (a peaked curve) or decentralized form the mode of both tails of the frequency curve (a flat topped curve). Hence the concepts of kurtosis and skewness are used in analyzing business data.

Usefulness of measures of skewness and kurtosis:

Skewness and kurtosis are two commonly listed values when you run a software's descriptive statistics function. Many books say that these two statistics give you insights into the shape of the distribution. Skewness is a measure of the symmetry in a distribution. A symmetrical dataset will have a skewness equal to 0. So, a normal distribution will have a skewness of 0. Skewness essentially measures the relative size of the two tails.

Kurtosis is a measure of the combined sizes of the two tails. It measures the amount of probability in the tails. The value is often compared to the kurtosis of the normal distribution, which is equal to 3. If the kurtosis is greater than 3, then the dataset has heavier tails than a normal distribution (more in the tails). If the kurtosis is less than 3, then the dataset has lighter tails than a normal distribution (less in the tails). Careful here. Kurtosis is sometimes reported as "excess kurtosis." Excess kurtosis is determined by subtracting 3 form the kurtosis. This makes the normal distribution kurtosis equal 0. Kurtosis originally was thought to measure the peakedness of a distribution. Though you will still see this as part of the definition in many places, this is a misconception. Skewness and kurtosis involve the tails of the distribution. These are presented in more detail below.

Distinction between Skewness and Kurtosis

Skewness	Kurtosis
(1) It measures the form of the data distribution.	(1) It measures the spread of the data distribution.
(2) It is zero for symmetric data distribution.	(2) It is normal for symmetric data distribution.

(3) It may be positive, negative or zero	(3) It may be platykurtic, mesokurtic or leptokurtic.
(4) It has three formulas for findings.	(4) It has two formulas for findings.
(5) It is widely used.	(5) Its uses are limited.

Distinction between Skewness and dispersion

Skewness	Dispersion
(1) It is relative measure.	(1) It is a absolute measure.
(2) It is not formatted.	(2) It is formatted.
(3) It is relatively difficult to make.	(3) It is relatively easier to make.
(4) It has limited option.	(4) It has many options.
(5) Its uses are limited.	(5) It is widely used.
(6) It is used to find the shape of the data distribution.	(6) It is used to compare among different data sets.

Problem 1.: You are given the following data pertaining to kilowatt hours of electricity consumed by 100 persons in Dhaka:

Consumption (In k-watt hours)	0-10	10-20	20-30	30-40	40-50
No. of users	6	25	36	20	13

Calculate: (i) Arithmetic mean; (ii) Standard deviation; (iii) Coefficient of skewness and write a commend.

Solution:

Calculation of A.M., S.D. and Sk_p

Marks	f	(X)	fX	fX ²
0 – 10	6	5	30	150
10 – 20	25	15	375	5625
20 – 30	36	25	900	22500
30 – 40	20	35	700	24500
40 – 50	13	45	585	26325
	N = 100		$\Sigma fX = 2590$	$\Sigma fX^2 = 79100$

$$(i) \text{ Mean} = \frac{\Sigma fX}{N} = \frac{2590}{100} = 25.9 \quad [\text{Ans.}]$$

$$(ii) \text{ S.D.} = \sqrt{\frac{\Sigma fX^2}{N} - \left(\frac{\Sigma fX}{N}\right)^2} = \sqrt{\frac{79100}{100} - \left(\frac{2590}{100}\right)^2} = 10.963 \quad [\text{Ans.}]$$

(iii) For highest frequency 36, mode lies in the class 20 – 30.

$$\text{Mode} = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times i = 20 + \frac{11}{11+16} \times 10 = 24.07$$

$$\therefore S_k_p = \frac{(\text{Mean} - \text{mode})}{\sigma} = \frac{(25.9 - 24.07)}{10.963} = 0.167 \quad [\text{Ans.}]$$

Problem 2

Find the first two central moments from the following data and hence find coefficient variation: (10, 5, 0, -4, -1, 6)

Solution:

Calculation of μ_1 and μ_2 and C.V.

X	$(X - \bar{X})$	$(X - \bar{X})^2$
10	7.33	53.73
5	2.33	5.43
0	- 2.67	7.13
- 4	- 6.67	44.49

- 1	- 3.67	13.47
6	3.35*	11.22
$\Sigma X = 16$	$\Sigma(X - \bar{X}) = 0$	$\Sigma(X - \bar{X})^2 = 135.47$

* Adjusted

$$\therefore \text{Mean } (\bar{X}) = \frac{\Sigma X}{N} = \frac{16}{6} = 2.67$$

$$(i) \text{ The first central moment } (\mu_1) = \frac{\Sigma(X - \bar{X})}{N} = \frac{0}{6} = 0 \quad [\text{Ans.}]$$

$$\text{The second central moment } (\mu_2) = \frac{\Sigma(X - \bar{X})^2}{N} = \frac{135.47}{6} = 22.58 \quad [\text{Ans.}]$$

(variance)

$$(ii) \text{ Here SD } (\sigma) = \sqrt{\frac{\Sigma(X - \bar{X})^2}{N}} = \sqrt{\frac{135.47}{6}} = 4.75$$

$$\therefore \text{C.V.} = \frac{\sigma}{\bar{X}} \times 100 = \frac{4.75}{2.67} \times 100 = 177.9\% \quad [\text{Ans.}]$$

Problem 3

Calculate coefficient of skewness of the above data (Given that standard deviation is 4.12). Based on the above result, what is your conclusion about the skewness?

Solution: We are given, Standard deviation (σ) = 4.12

$$\text{Mean} = 30.09$$

$$\text{Mode} = 34$$

$$\therefore \text{Coefficient of skewness } (Sk_p) = \frac{\text{Mean} - \text{Mode}}{\sigma} = \frac{30.09 - 34}{4.12} = -0.949 \quad [\text{Ans.}]$$

Problem 4

Find coefficient of skewness from the following data:

25 -5 18 30 16 29

Solution: Given

- 16 18 25 29 30
5

$$\text{Mean } (\bar{X}) = \frac{\sum X}{n} = \frac{-5+16+18+25+29+30}{6} = 18.83$$

$$\text{Median (Md)} = \text{Mid-value} = (18+25)/2 = 21.5$$

$$\text{Mode (Mo)} = 3(\text{median}) - 2(\text{mean}) = 3(21.5) - 2(18.83) = 26.84$$

$$\begin{aligned} \text{SD } (\sigma) &= \sqrt{\frac{\sum (X - \bar{X})^2}{n}} \\ &= \sqrt{\frac{(-5-18.83)^2 + (16-18.83)^2 + (18-18.83)^2 + (25-18.83)^2 + (29-18.83)^2 + (30-18.83)^2}{6}} \\ &= 11.85 \end{aligned}$$

$$\therefore \text{Coefficient of skewness } (Sk_p) = \frac{\text{Mean} - \text{Mode}}{SD} = \frac{18.83 - 26.84}{11.85} = -0.68 \text{ [Ans]}$$

Problem 5

Calculate the first four moments about mean from the age of five labourers:

Age (in year) : 25, 27, 30, 31, 33.

Solution:

Calculation of Central Moments

X	$(X - \bar{X})$	$(X - \bar{X})^2$	$(X - \bar{X})^3$	$(X - \bar{X})^4$
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25	- 4.2	17.64	- 74.088	311.1696
27	-2.2	4.84	- 10.648	23.4256
30	0.8	0.64	0.512	0.4096
31	1.8	3.24	5.832	10.4976
33	3.8	14.44	54.872	208.5136
ΣX = 146	$\Sigma(X - \bar{X})$ = 0	$(X - \bar{X})^2$ = 40.8	$(X - \bar{X})^3$ = - 23.52	$(X - \bar{X})^4$ = 554.016

Here $\bar{X} = \frac{\Sigma X}{n} = \frac{146}{5} = 29.2$

∴ The first four moments about mean (first four central moments) are

$$\mu_1 = \frac{\Sigma (X - \bar{X})}{n} = \frac{0}{5} = 0 \quad [\text{Ans.}]$$

$$\mu_2 = \frac{\Sigma (X - \bar{X})^2}{n} = \frac{40.8}{5} = 8.16 \quad [\text{Ans.}]$$

$$\mu_3 = \frac{\Sigma (X - \bar{X})^3}{n} = \frac{-23.52}{5} = -4.704 \quad [\text{Ans.}]$$

$$\mu_4 = \frac{\Sigma (X - \bar{X})^4}{n} = \frac{554.016}{5} = 110.8032 \quad [\text{Ans.}]$$

